

# Wireshark + OT Protocol Analysis

## Modbus/TCP Traffic Baseline, Packet Inspection, and Anomaly Check

Capture file: ModbusTCP.pcap | Tool: Wireshark

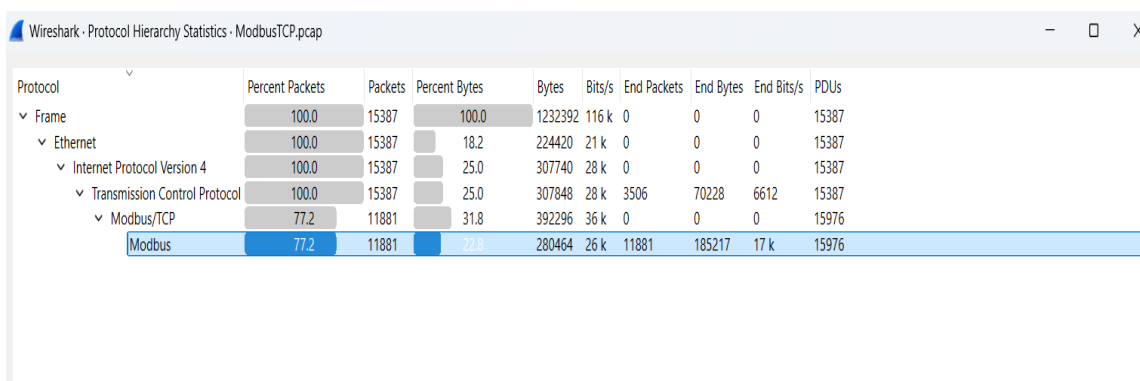
Date: June 2026

## 1. Project Overview

This report covers the analysis of a Modbus/TCP packet capture (ModbusTCP.pcap) using Wireshark, as part of an OT/ICS security exercise. The capture is used to establish a baseline of normal network behavior, inspect write commands at the packet level, and check for anomalous activity.

## 2. Protocol Hierarchy

Protocol Hierarchy statistics were used to confirm the capture contains genuine Modbus/TCP traffic.



| Protocol                      | Percent Packets | Packets | Percent Bytes | Bytes   | Bits/s | End Packets | End Bytes | End Bits/s | PDU/s |
|-------------------------------|-----------------|---------|---------------|---------|--------|-------------|-----------|------------|-------|
| Frame                         | 100.0           | 15387   | 100.0         | 1232392 | 116 k  | 0           | 0         | 0          | 15387 |
| Ethernet                      | 100.0           | 15387   | 18.2          | 224420  | 21 k   | 0           | 0         | 0          | 15387 |
| Internet Protocol Version 4   | 100.0           | 15387   | 25.0          | 307740  | 28 k   | 0           | 0         | 0          | 15387 |
| Transmission Control Protocol | 100.0           | 15387   | 25.0          | 307848  | 28 k   | 3506        | 70228     | 6612       | 15387 |
| Modbus/TCP                    | 77.2            | 11881   | 31.8          | 392296  | 36 k   | 0           | 0         | 0          | 15976 |
| Modbus                        | 77.2            | 11881   | 31.8          | 280464  | 26 k   | 11881       | 185217    | 17 k       | 15976 |

Fig. 1: Protocol Hierarchy confirms 11,881 Modbus/TCP packets (77.2% of 15,387 total frames), with multiple PDUs per packet (15,976) indicating active polling. The remainder is TCP connection overhead.

## 3. Master and Slave Identification

The TCP Conversations view was used to identify the devices on the network and determine master/slave roles.

| Ethernet - 13 |        | IPv4 - 13    |        | IPv6    |       | TCP - 14  |               | UDP              |               |             |               |             |           |          |              |              |
|---------------|--------|--------------|--------|---------|-------|-----------|---------------|------------------|---------------|-------------|---------------|-------------|-----------|----------|--------------|--------------|
| Address A     | Port A | Address B    | Port B | Packets | Bytes | Stream ID | Total Packets | Percent Filtered | Packets A → B | Bytes A → B | Packets B → A | Bytes B → A | Rel Start | Duration | Bits/s A → B | Bits/s B → A |
| 141.81.0.10   | 64338  | 141.81.0.24  | 502    | 114     | 8 kB  | 1         | 1,326         | 8.60%            | 114           | 8 kB        | 0             | 0 bytes     | 0.072663  | 84.8306  | 731 bits/s   | 0 bits/s     |
| 141.81.0.10   | 51411  | 141.81.0.26  | 502    | 163     | 12 kB | 6         | 987           | 16.51%           | 163           | 12 kB       | 0             | 0 bytes     | 0.127740  | 84.7256  | 1115 bits/s  | 0 bits/s     |
| 141.81.0.10   | 53414  | 141.81.0.44  | 502    | 118     | 8 kB  | 2         | 1,207         | 9.78%            | 118           | 8 kB        | 0             | 0 bytes     | 0.082747  | 84.8105  | 769 bits/s   | 0 bits/s     |
| 141.81.0.10   | 59758  | 141.81.0.46  | 502    | 90      | 7 kB  | 8         | 553           | 16.27%           | 90            | 7 kB        | 0             | 0 bytes     | 0.179846  | 55.6095  | 1007 bits/s  | 0 bits/s     |
| 141.81.0.10   | 59796  | 141.81.0.46  | 502    | 38      | 3 kB  | 13        | 277           | 13.72%           | 38            | 3 kB        | 0             | 0 bytes     | 56.010762 | 28.8926  | 738 bits/s   | 0 bits/s     |
| 141.81.0.10   | 64368  | 141.81.0.64  | 502    | 114     | 8 kB  | 9         | 1,222         | 9.33%            | 114           | 8 kB        | 0             | 0 bytes     | 0.260110  | 84.6232  | 739 bits/s   | 0 bits/s     |
| 141.81.0.10   | 54138  | 141.81.0.66  | 502    | 160     | 12 kB | 7         | 1,555         | 10.29%           | 160           | 12 kB       | 0             | 0 bytes     | 0.173836  | 84.6450  | 1124 bits/s  | 0 bits/s     |
| 141.81.0.10   | 50594  | 141.81.0.84  | 502    | 116     | 8 kB  | 10        | 1,265         | 9.17%            | 116           | 8 kB        | 0             | 0 bytes     | 0.296096  | 84.6173  | 748 bits/s   | 0 bits/s     |
| 141.81.0.10   | 64340  | 141.81.0.104 | 502    | 117     | 8 kB  | 3         | 1,204         | 9.72%            | 117           | 8 kB        | 0             | 0 bytes     | 0.082881  | 84.8753  | 749 bits/s   | 0 bits/s     |
| 141.81.0.10   | 59599  | 141.81.0.143 | 502    | 175     | 13 kB | 11        | 1,122         | 15.60%           | 175           | 13 kB       | 0             | 0 bytes     | 0.353049  | 84.5383  | 1277 bits/s  | 0 bits/s     |
| 141.81.0.10   | 64341  | 141.81.0.144 | 502    | 115     | 8 kB  | 4         | 949           | 12.12%           | 115           | 8 kB        | 0             | 0 bytes     | 0.082977  | 84.8752  | 737 bits/s   | 0 bits/s     |
| 141.81.0.10   | 59598  | 141.81.0.163 | 502    | 175     | 14 kB | 12        | 1,248         | 14.02%           | 175           | 14 kB       | 0             | 0 bytes     | 0.429800  | 84.4844  | 1284 bits/s  | 0 bits/s     |
| 141.81.0.10   | 64342  | 141.81.0.164 | 502    | 116     | 8 kB  | 5         | 954           | 12.16%           | 116           | 8 kB        | 0             | 0 bytes     | 0.083193  | 84.8753  | 743 bits/s   | 0 bits/s     |
| 141.81.0.86   | 502    | 141.81.0.10  | 57184  | 182     | 13 kB | 0         | 1,518         | 11.99%           | 0             | 0 bytes     | 182           | 13 kB       | 0.000000  | 84.9284  | 0 bits/s     | 1229 bits/s  |

Fig. 2: A single master, 141.81.0.10, communicates with 13 slave devices, all on standard Modbus port 502.

- Slaves (last octet): 24, 26, 44, 46, 64, 66, 84, 86, 104, 143, 144, 163, 164.
- Master uses ephemeral/random source ports per session; slaves consistently use fixed port 502.

### 3.1 Query / Response Verification

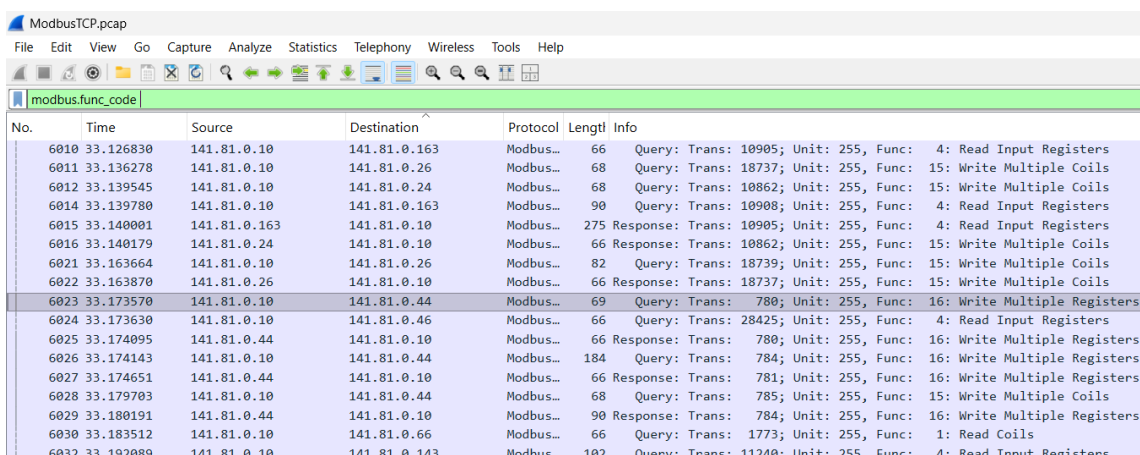
Master/slave roles were confirmed independently of port numbers using Wireshark's own Query/Response labelling.

|            |             |             |           |     |                                                                  |
|------------|-------------|-------------|-----------|-----|------------------------------------------------------------------|
| 4 0.001032 | 141.81.0.10 | 141.81.0.86 | Modbus... | 66  | Query: Trans: 1; Unit: 255, Func: 2: Read Discrete Inputs        |
| 5 0.048116 | 141.81.0.86 | 141.81.0.10 | Modbus... | 80  | Response: Trans: 1; Unit: 255, Func: 2: Read Discrete Inputs     |
| 6 0.072663 | 141.81.0.10 | 141.81.0.24 | Modbus... | 66  | Query: Trans: 10613; Unit: 255, Func: 4: Read Input Registers    |
| 7 0.073315 | 141.81.0.24 | 141.81.0.10 | Modbus... | 143 | Response: Trans: 10613; Unit: 255, Func: 4: Read Input Registers |

Fig. 3: Every packet from 141.81.0.10 is labelled “Query”, and every reply from a slave is labelled “Response”, confirming roles at the protocol level.

## 4. Function Code Distribution

All Modbus packets were filtered by function code to characterize normal read/write behavior.



| No.  | Time      | Source       | Destination  | Protocol  | Length | Info                                                                |
|------|-----------|--------------|--------------|-----------|--------|---------------------------------------------------------------------|
| 6010 | 33.126830 | 141.81.0.10  | 141.81.0.163 | Modbus... | 66     | Query: Trans: 10905; Unit: 255, Func: 4: Read Input Registers       |
| 6011 | 33.136278 | 141.81.0.10  | 141.81.0.26  | Modbus... | 68     | Query: Trans: 18737; Unit: 255, Func: 15: Write Multiple Coils      |
| 6012 | 33.139545 | 141.81.0.10  | 141.81.0.24  | Modbus... | 68     | Query: Trans: 10862; Unit: 255, Func: 15: Write Multiple Coils      |
| 6014 | 33.139780 | 141.81.0.10  | 141.81.0.163 | Modbus... | 90     | Query: Trans: 10908; Unit: 255, Func: 4: Read Input Registers       |
| 6015 | 33.140001 | 141.81.0.163 | 141.81.0.10  | Modbus... | 275    | Response: Trans: 10905; Unit: 255, Func: 4: Read Input Registers    |
| 6016 | 33.140179 | 141.81.0.24  | 141.81.0.10  | Modbus... | 66     | Response: Trans: 10862; Unit: 255, Func: 15: Write Multiple Coils   |
| 6021 | 33.163664 | 141.81.0.10  | 141.81.0.26  | Modbus... | 82     | Query: Trans: 18739; Unit: 255, Func: 15: Write Multiple Coils      |
| 6022 | 33.163870 | 141.81.0.26  | 141.81.0.10  | Modbus... | 66     | Response: Trans: 18737; Unit: 255, Func: 15: Write Multiple Coils   |
| 6023 | 33.173570 | 141.81.0.10  | 141.81.0.44  | Modbus... | 69     | Query: Trans: 780; Unit: 255, Func: 16: Write Multiple Registers    |
| 6024 | 33.173630 | 141.81.0.10  | 141.81.0.46  | Modbus... | 66     | Query: Trans: 28425; Unit: 255, Func: 4: Read Input Registers       |
| 6025 | 33.174095 | 141.81.0.44  | 141.81.0.10  | Modbus... | 66     | Response: Trans: 780; Unit: 255, Func: 16: Write Multiple Registers |
| 6026 | 33.174143 | 141.81.0.10  | 141.81.0.44  | Modbus... | 184    | Query: Trans: 784; Unit: 255, Func: 16: Write Multiple Registers    |
| 6027 | 33.174651 | 141.81.0.44  | 141.81.0.10  | Modbus... | 66     | Response: Trans: 781; Unit: 255, Func: 16: Write Multiple Registers |
| 6028 | 33.179703 | 141.81.0.10  | 141.81.0.44  | Modbus... | 68     | Query: Trans: 785; Unit: 255, Func: 15: Write Multiple Coils        |
| 6029 | 33.180191 | 141.81.0.44  | 141.81.0.10  | Modbus... | 90     | Response: Trans: 784; Unit: 255, Func: 16: Write Multiple Registers |
| 6030 | 33.183512 | 141.81.0.10  | 141.81.0.66  | Modbus... | 66     | Query: Trans: 1773; Unit: 255, Func: 1: Read Coils                  |
| 6032 | 33.192089 | 141.81.0.10  | 141.81.0.143 | Modbus... | 102    | Query: Trans: 11240; Unit: 255, Func: 4: Read Input Registers       |

Fig. 4: Reads (Read Coils, Read Discrete Inputs, Read Input Registers) dominate the capture, consistent with SCADA-style polling. Two write types appear: Write Multiple Coils (15) and Write Multiple Registers (16).

| Function Code | Name                     | Type  |
|---------------|--------------------------|-------|
| 01            | Read Coils               | Read  |
| 02            | Read Discrete Inputs     | Read  |
| 04            | Read Input Registers     | Read  |
| 15 (0x0F)     | Write Multiple Coils     | Write |
| 16 (0x10)     | Write Multiple Registers | Write |

## 5. Write Command Baseline

Write function codes (15, 16) were filtered to commands directed at slaves (tcp.dstport == 502), excluding acknowledgement responses which echo the same function code back to the master.

| No.   | Time      | Source      | Destination  | Protocol   | Length | Info                                                           |
|-------|-----------|-------------|--------------|------------|--------|----------------------------------------------------------------|
| 15366 | 84.839240 | 141.81.0.10 | 141.81.0.163 | Modbus/TCP | 106    | Query: Trans: 11307; Unit: 255, Func: 2: Read Discrete Inputs  |
| 15363 | 84.816395 | 141.81.0.10 | 141.81.0.143 | Modbus/TCP | 82     | Query: Trans: 11638; Unit: 255, Func: 15: Write Multiple Coils |
| 15359 | 84.800799 | 141.81.0.10 | 141.81.0.164 | Modbus/TCP | 68     | Query: Trans: 13020; Unit: 255, Func: 15: Write Multiple Coils |
| 15358 | 84.797356 | 141.81.0.10 | 141.81.0.163 | Modbus/TCP | 69     | Query: Trans: 11303; Unit: 255, Func: 15: Write Multiple Coils |
| 15357 | 84.769836 | 141.81.0.10 | 141.81.0.66  | Modbus/TCP | 80     | Query: Trans: 2308; Unit: 255, Func: 2: Read Discrete Inputs   |
| 15347 | 84.759133 | 141.81.0.10 | 141.81.0.164 | Modbus/TCP | 68     | Query: Trans: 13019; Unit: 255, Func: 15: Write Multiple Coils |
| 15346 | 84.759013 | 141.81.0.10 | 141.81.0.144 | Modbus/TCP | 68     | Query: Trans: 12960; Unit: 255, Func: 15: Write Multiple Coils |
| 15345 | 84.757060 | 141.81.0.10 | 141.81.0.163 | Modbus/TCP | 68     | Query: Trans: 11302; Unit: 255, Func: 15: Write Multiple Coils |
| 15344 | 84.756735 | 141.81.0.10 | 141.81.0.143 | Modbus/TCP | 69     | Query: Trans: 11636; Unit: 255, Func: 15: Write Multiple Coils |
| 15342 | 84.723924 | 141.81.0.10 | 141.81.0.104 | Modbus/TCP | 68     | Query: Trans: 20881; Unit: 255, Func: 15: Write Multiple Coils |
| 15340 | 84.722865 | 141.81.0.10 | 141.81.0.104 | Modbus/TCP | 68     | Query: Trans: 20880; Unit: 255, Func: 15: Write Multiple Coils |
| 15338 | 84.722887 | 141.81.0.10 | 141.81.0.84  | Modbus/TCP | 68     | Query: Trans: 7152; Unit: 255, Func: 15: Write Multiple Coils  |
| 15337 | 84.720863 | 141.81.0.10 | 141.81.0.66  | Modbus/TCP | 68     | Query: Trans: 2306; Unit: 255, Func: 15: Write Multiple Coils  |
| 15332 | 84.715811 | 141.81.0.10 | 141.81.0.86  | Modbus/TCP | 82     | Query: Trans: 881; Unit: 255, Func: 15: Write Multiple Coils   |
| 15314 | 84.685873 | 141.81.0.10 | 141.81.0.84  | Modbus/TCP | 68     | Query: Trans: 7151; Unit: 255, Func: 15: Write Multiple Coils  |
| 15312 | 84.684970 | 141.81.0.10 | 141.81.0.64  | Modbus/TCP | 68     | Query: Trans: 12649; Unit: 255, Func: 15: Write Multiple Coils |
| 15311 | 84.678359 | 141.81.0.10 | 141.81.0.86  | Modbus/TCP | 68     | Query: Trans: 879; Unit: 255, Func: 15: Write Multiple Coils   |

Fig. 5: 1,793 write operations were found in total, all from the master (141.81.0.10) and targeting only the 13 known slaves - no unexpected source or destination IPs.

## 6. Baseline Summary

Traffic follows a single, consistent pattern: one master polling 13 slaves over Modbus/TCP, with reads far outnumbering writes, as expected for SCADA-style polling. No anomalous source IPs, destination IPs, or out-of-pattern behavior were observed at this stage.

## 7. Packet-Level Field Inspection

### 7.1 Sample Write Packets

Individual write packets were opened at the field level to record the exact register/coil address and value written, using:

```
(modbus.func_code==15 || modbus.func_code==16) && tcp.dstport==502
```

|            |                                                          |
|------------|----------------------------------------------------------|
| Modbus/TCP | Transaction Identifier: 780                              |
|            | Protocol Identifier: 0                                   |
|            | Length: 9                                                |
|            | Unit Identifier: 255                                     |
| Modbus     | 0... .... = Exception: No                                |
|            | .001 0000 = Function Code: Write Multiple Registers (16) |
|            | Reference Number: 2100                                   |
|            | Word Count: 1                                            |
|            | Byte Count: 2                                            |
|            | > Register 2100 (UINT16): 3                              |

Fig. 6: Example write packet - Function Code 16 (Write Multiple Registers), Reference Number 2100, value 3. A sample of 10 write packets across 7 slave devices was recorded:

| #  | Source -> Dest      | Func | Type            | Address | Value   |
|----|---------------------|------|-----------------|---------|---------|
| 1  | 141.81.0.10 -> .163 | 15   | Single Coil     | 8       | 1 (ON)  |
| 2  | 141.81.0.10 -> .163 | 15   | Single Coil     | 6       | 0 (OFF) |
| 3  | 141.81.0.10 -> .163 | 15   | Multi Coil (3)  | 7-9     | All 0   |
| 4  | 141.81.0.10 -> .86  | 16   | Register        | 2100    | 3       |
| 5  | 141.81.0.10 -> .104 | 15   | Single Coil     | 5       | 0 (OFF) |
| 6  | 141.81.0.10 -> .44  | 16   | Register        | 2100    | 3       |
| 7  | 141.81.0.10 -> .143 | 15   | Multi Coil (10) | 9-18    | -       |
| 8  | 141.81.0.10 -> .46  | 15   | Single Coil     | 0       | 1 (ON)  |
| 9  | 141.81.0.10 -> .164 | 15   | Single Coil     | 5       | 0 (OFF) |
| 10 | 141.81.0.10 -> .163 | 15   | Multi Coil (10) | 9-18    | -       |

## 7.2 Full-Range Check

This 10-packet sample initially suggested coil addresses stayed within 0-18 and register writes always targeted address 2100 with value 3. A full scan of all 1,793 write packets, sorted by Reference Number, showed this was incomplete: addresses ranged from 0 to 2,258, forming two clusters rather than one fixed value.

| No.   | Time      | Source      | Destination | Protocol   | Length | Reference Number   | Info                                                           |
|-------|-----------|-------------|-------------|------------|--------|--------------------|----------------------------------------------------------------|
| 14261 | 76.775676 | 141.81.0.10 | 141.81.0.66 | Modbus/TCP | 92     | 2258,5,99          | Query: Trans: 2246; Unit: 255, Func: 2: Read Discrete Inputs   |
| 11582 | 64.788522 | 141.81.0.10 | 141.81.0.66 | Modbus/TCP | 92     | 2258,5,99          | Query: Trans: 2100; Unit: 255, Func: 2: Read Discrete Inputs   |
| 6067  | 33.264435 | 141.81.0.10 | 141.81.0.66 | Modbus/TCP | 92     | 2258,5,99          | Query: Trans: 28433; Unit: 255, Func: 2: Read Discrete Inputs  |
| 8448  | 45.714107 | 141.81.0.10 | 141.81.0.66 | Modbus/TCP | 94     | 205,7,5            | Query: Trans: 28544; Unit: 255, Func: 15: Write Multiple Coils |
| 15126 | 83.770032 | 141.81.0.10 | 141.81.0.66 | Modbus/TCP | 80     | 99,5               | Query: Trans: 2298; Unit: 255, Func: 15: Write Multiple Coils  |
| 12398 | 68.784663 | 141.81.0.10 | 141.81.0.66 | Modbus/TCP | 80     | 99,5               | Query: Trans: 2142; Unit: 255, Func: 15: Write Multiple Coils  |
| 2385  | 13.037820 | 141.81.0.10 | 141.81.0.66 | Modbus/TCP | 80     | 99,5               | Query: Trans: 130; Unit: 255, Func: 15: Write Multiple Coils   |
| 216   | 1.049560  | 141.81.0.10 | 141.81.0.66 | Modbus/TCP | 80     | 99,5               | Query: Trans: 13; Unit: 255, Func: 15: Write Multiple Coils    |
| 14628 | 80.772959 | 141.81.0.10 | 141.81.0.66 | Modbus/TCP | 94     | 7,99,5             | Query: Trans: 2267; Unit: 255, Func: 15: Write Multiple Coils  |
| 9589  | 52.195093 | 141.81.0.10 | 141.81.0.66 | Modbus/TCP | 130    | 7,5,399,1,41,22... | Query: Trans: 18862; Unit: 255, Func: 4: Read Input Registers  |

| No.   | Time      | Source      | Destination | Protocol   | Length | Reference Number | Info                                                           |
|-------|-----------|-------------|-------------|------------|--------|------------------|----------------------------------------------------------------|
| 4645  | 25.273807 | 141.81.0.10 | 141.81.0.66 | Modbus/TCP | 92     | 0,0,5            | Query: Trans: 268; Unit: 255, Func: 15: Write Multiple Coils   |
| 18138 | 55.296088 | 141.81.0.10 | 141.81.0.66 | Modbus/TCP | 92     | 0,0,5            | Query: Trans: 583; Unit: 255, Func: 15: Write Multiple Coils   |
| 1706  | 9.192992  | 141.81.0.10 | 141.81.0.66 | Modbus/TCP | 106    | 0,0,7,5          | Query: Trans: 1523; Unit: 255, Func: 15: Write Multiple Coils  |
| 2055  | 11.191846 | 141.81.0.10 | 141.81.0.66 | Modbus/TCP | 106    | 0,0,7,5          | Query: Trans: 1544; Unit: 255, Func: 15: Write Multiple Coils  |
| 7215  | 39.214404 | 141.81.0.10 | 141.81.0.66 | Modbus/TCP | 106    | 0,0,7,5          | Query: Trans: 1839; Unit: 255, Func: 15: Write Multiple Coils  |
| 1172  | 6.293977  | 141.81.0.10 | 141.81.0.66 | Modbus/TCP | 80     | 0,5              | Query: Trans: 67; Unit: 255, Func: 15: Write Multiple Coils    |
| 83    | 0.299908  | 141.81.0.10 | 141.81.0.66 | Modbus/TCP | 94     | 0,7,5            | Query: Trans: 5; Unit: 255, Func: 15: Write Multiple Coils     |
| 577   | 3.198907  | 141.81.0.10 | 141.81.0.66 | Modbus/TCP | 94     | 0,7,5            | Query: Trans: 1459; Unit: 255, Func: 15: Write Multiple Coils  |
| 2759  | 15.187126 | 141.81.0.10 | 141.81.0.66 | Modbus/TCP | 94     | 0,7,5            | Query: Trans: 1586; Unit: 255, Func: 15: Write Multiple Coils  |
| 5532  | 30.223250 | 141.81.0.10 | 141.81.0.66 | Modbus/TCP | 94     | 0,7,5            | Query: Trans: 1745; Unit: 255, Func: 15: Write Multiple Coils  |
| 6049  | 33.220251 | 141.81.0.10 | 141.81.0.66 | Modbus/TCP | 94     | 0,7,5            | Query: Trans: 1776; Unit: 255, Func: 15: Write Multiple Coils  |
| 6444  | 35.217424 | 141.81.0.10 | 141.81.0.66 | Modbus/TCP | 94     | 0,7,5            | Query: Trans: 1797; Unit: 255, Func: 15: Write Multiple Coils  |
| 6728  | 36.460634 | 141.81.0.10 | 141.81.0.66 | Modbus/TCP | 94     | 0,7,5            | Query: Trans: 18765; Unit: 255, Func: 15: Write Multiple Coils |

Fig. 7: Sorting write packets by Reference Number shows the low end (top) and high end (bottom) of the range.

- Low cluster (~0-400): coil addresses and small register addresses, e.g. 0, 1, 2, 5, 7, 41, 99, 205, 399.
- High cluster (~1,100-2,258): register addresses, e.g. 1100, 1300, 2100-2105, 2200, 2219, 2258.

The gap between the two clusters (400-1,100) is empty for write traffic specifically. It first looked populated because a single TCP packet often bundles several Modbus PDUs together, so a packet matching the write filter can show a long list of Reference Numbers (e.g. 5, 399, 1, 41, 2219) that actually belong to bundled read PDUs travelling alongside the write in the same segment.

| No.   | Time      | Source      | Destination  | Protocol   | Length | Reference Number   | Info                                                          |
|-------|-----------|-------------|--------------|------------|--------|--------------------|---------------------------------------------------------------|
| 12167 | 67.587447 | 141.81.0.10 | 141.81.0.143 | Modbus/TCP | 104    | 0,0,699,99         | Query: Trans: 11506; Unit: 255, Func: 2: Read Discrete Inputs |
| 15098 | 83.678840 | 141.81.0.10 | 141.81.0.66  | Modbus/TCP | 80     | 0,2258             | Query: Trans: 2295; Unit: 255, Func: 4: Read Input Registers  |
| 14230 | 78.674966 | 141.81.0.10 | 141.81.0.66  | Modbus/TCP | 104    | 1,0,41,2219        | Query: Trans: 2242; Unit: 255, Func: 4: Read Input Registers  |
| 13680 | 75.677927 | 141.81.0.10 | 141.81.0.66  | Modbus/TCP | 118    | 1,7,5,41,2219      | Query: Trans: 2232; Unit: 255, Func: 4: Read Input Registers  |
| 1896  | 10.297873 | 141.81.0.10 | 141.81.0.64  | Modbus/TCP | 104    | 5,1100,1300,203    | Query: Trans: 12124; Unit: 255, Func: 2: Read Discrete Inputs |
| 5881  | 32.298567 | 141.81.0.10 | 141.81.0.64  | Modbus/TCP | 104    | 5,1100,1300,203    | Query: Trans: 12271; Unit: 255, Func: 2: Read Discrete Inputs |
| 6492  | 35.360566 | 141.81.0.10 | 141.81.0.46  | Modbus/TCP | 92     | 5,2219,2258        | Query: Trans: 28453; Unit: 255, Func: 4: Read Input Registers |
| 370   | 2.192659  | 141.81.0.10 | 141.81.0.26  | Modbus/TCP | 116    | 5,399,1,41,2219    | Query: Trans: 18540; Unit: 255, Func: 4: Read Input Registers |
| 10292 | 56.189060 | 141.81.0.10 | 141.81.0.26  | Modbus/TCP | 116    | 5,399,1,41,2219    | Query: Trans: 18887; Unit: 255, Func: 4: Read Input Registers |
| 13277 | 73.680717 | 141.81.0.10 | 141.81.0.66  | Modbus/TCP | 116    | 5,399,1,41,2219    | Query: Trans: 2192; Unit: 255, Func: 4: Read Input Registers  |
| 1325  | 6.941699  | 141.81.0.10 | 141.81.0.86  | Modbus/TCP | 128    | 5,399,1,41,2219... | Query: Trans: 75; Unit: 255, Func: 4: Read Input Registers    |
| 3140  | 17.177496 | 141.81.0.10 | 141.81.0.46  | Modbus/TCP | 128    | 5,399,1,41,2219... | Query: Trans: 28327; Unit: 255, Func: 4: Read Input Registers |
| 9905  | 54.191300 | 141.81.0.10 | 141.81.0.26  | Modbus/TCP | 128    | 5,399,1,41,2219... | Query: Trans: 18875; Unit: 255, Func: 4: Read Input Registers |
| 3500  | 18.985401 | 141.81.0.10 | 141.81.0.86  | Modbus/TCP | 80     | 7,2258             | Query: Trans: 201; Unit: 255, Func: 4: Read Input Registers   |
| 4996  | 27.240341 | 141.81.0.10 | 141.81.0.46  | Modbus/TCP | 94     | 7,5,2258           | Query: Trans: 28392; Unit: 255, Func: 4: Read Input Registers |
| 9509  | 52.195093 | 141.81.0.10 | 141.81.0.26  | Modbus/TCP | 130    | 7,5,399,1,41,22... | Query: Trans: 18862; Unit: 255, Func: 4: Read Input Registers |
| 15366 | 84.839240 | 141.81.0.10 | 141.81.0.163 | Modbus/TCP | 106    | 8,6,699,99         | Query: Trans: 11307; Unit: 255, Func: 2: Read Discrete Inputs |

Fig 8: Analysis of Packet #370. This packet appears in the write-filtered results due to TCP bundling. Expanding the packet metadata reveals that Reference Numbers 5, 399, 1, 41, and 2219 are associated with Read (FC2) PDUs, while the Write operation (FC15) within the same segment targets authorized addresses.

## 8. Timing Check

Write packets were re-examined using View -> Time Display Format -> Seconds Since Previous Displayed Packet, to check for burst patterns that might indicate scripted or malicious traffic rather than normal polling.

| No.  | Time     | Source      |
|------|----------|-------------|
| 3530 | 0.031644 | 141.81.0.10 |
| 36   | 0.002825 | 141.81.0.10 |
| 164  | 0.417050 | 141.81.0.10 |
| 165  | 0.001733 | 141.81.0.10 |
| 373  | 0.000056 | 141.81.0.10 |
| 388  | 0.000386 | 141.81.0.10 |
| 419  | 0.010169 | 141.81.0.10 |
| 422  | 0.000709 | 141.81.0.10 |
| 430  | 0.005130 | 141.81.0.10 |
| 442  | 0.003425 | 141.81.0.10 |
| 445  | 0.004413 | 141.81.0.10 |
| 541  | 0.650777 | 141.81.0.10 |
| 542  | 0.001805 | 141.81.0.10 |
| 551  | 0.003640 | 141.81.0.10 |
| 552  | 0.005529 | 141.81.0.10 |
| 722  | 0.000546 | 141.81.0.10 |
| 733  | 0.007556 | 141.81.0.10 |

Fig. 9: Time-since-previous-packet values for consecutive write packets.

Gaps between consecutive write packets range from about 0.0003 seconds to 0.65 seconds, with no fixed interval and no burst of many writes within milliseconds of each other. This is consistent with normal, asynchronous polling rather than a scripted or automated injection.

## 9. Anomaly Injection and Detection Validation

Since the real capture showed no anomaly, the detection approach was validated by injecting a synthetic unauthorized write into a copy of the capture, then confirming it is caught both manually (Wireshark) and automatically (Suricata).

### 9.1 Injecting a Synthetic Anomaly

A known write packet was cloned with Scapy, its source IP changed from the master (141.81.0.10) to an unauthorized address (192.168.50.99), and checksums recalculated so the packet remains well-formed:

```
from scapy.all import rdpcap, wrpcap, IP, TCP
```

```
packets = rdpcap("ModbusTCP.pcap")
malicious = packets[34].copy() # a known Write Multiple Coils packet
```

```

malicious[IP].src = "192.168.50.99"      # unauthorized source
del malicious[IP].chksum
if malicious.haslayer(TCP):
    del malicious[TCP].chksum
packets.append(malicious)
wrpcap("ModbusTCP_injected.pcap", packets)

```

The same filter used throughout this report was then run on both files:

```

(modbus.func_code==15 || modbus.func_code==16) && tcp.dstport==502 && ip.src !=
141.81.0.10

```

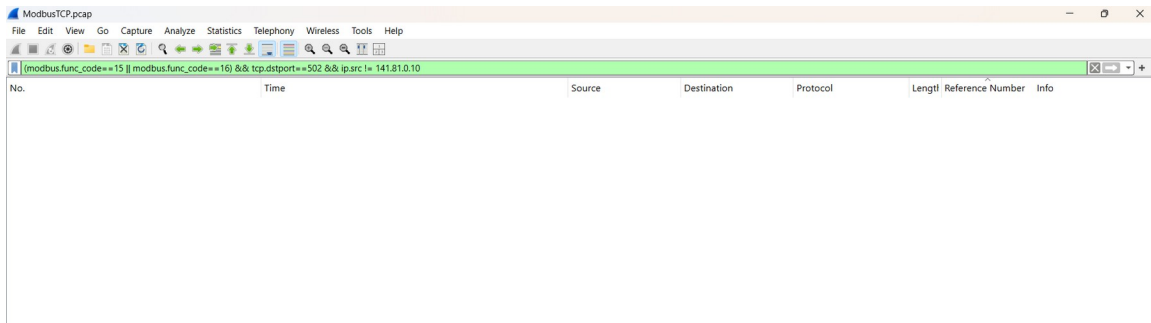


Fig. 10: Original capture - zero results. Every write comes from the master.

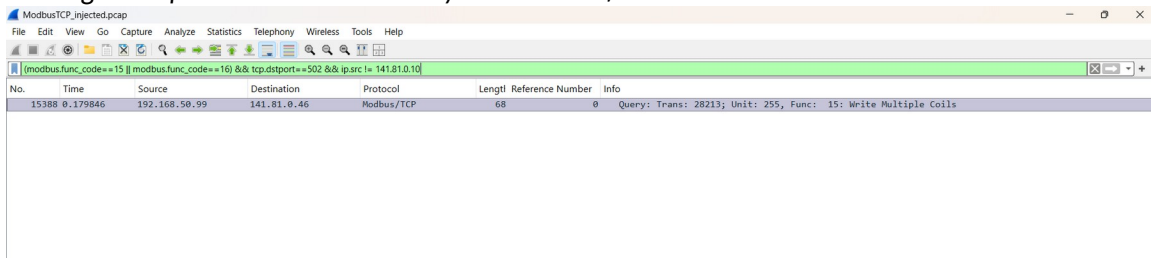


Fig. 11: Injected capture - one result, the synthetic write from 192.168.50.99 to slave 141.81.0.46.

## 9.2 Detection Rule (Suricata)

The same logic was encoded as a Suricata rule, so detection does not depend on someone manually applying a filter. The rule flags traffic to port 502 where the source is not the known master and the function code - located at byte offset 7 of the Modbus/TCP header - is 0x0F or 0x10, the two write types observed in this capture:

```

Bytes 0-1 -> Transaction ID
Bytes 2-3 -> Protocol ID
Bytes 4-5 -> Length
Byte 6 -> Unit ID
Byte 7 -> Function Code
Bytes 8-9 -> Reference Number (Starting Address)
Bytes 10-11 -> Quantity

```

Fig. 12: Modbus/TCP (MBAP) header layout. The function code sits at byte offset 7, which the rule checks directly.

```

alert tcp !141.81.0.10 any -> any 502 (msg:"Unauthorized Write Multiple Coils";
content:"|0F|"; offset:7; depth:1; sid:1000003; rev:1;)
alert tcp !141.81.0.10 any -> any 502 (msg:"Unauthorized Write Multiple
Registers"; content:"|10|"; offset:7; depth:1; sid:1000004; rev:1;)

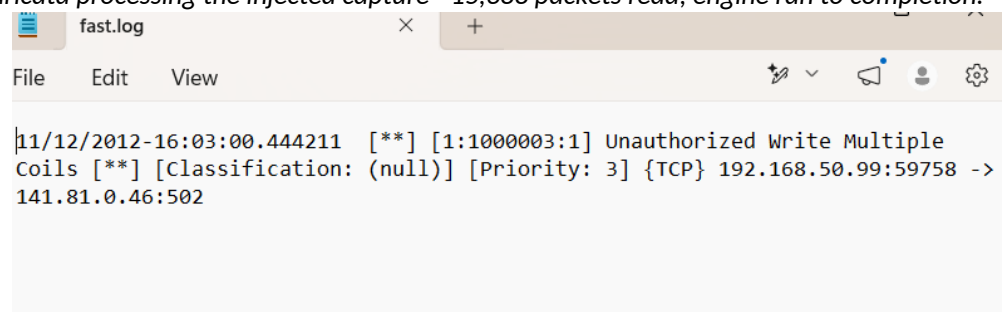
```

This ruleset covers only the write types seen in this capture. A production deployment would also cover Function Codes 05 and 06 (single coil/register writes), even though they were not observed here.

```
install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

C:\WINDOWS\system32> cd "C:\Users\Sheheryar Altaf\Desktop\project"
C:\Users\Sheheryar Altaf\Desktop\project> & "C:\Program Files\Suricata\suricata.exe" -r ModbusTCP_injected.pcap -S "
Program Files\Suricata\rules\modbus.rules" -l .\logs
info: win32-service: Running as service: no
suricata: This is Suricata version 8.0.5 RELEASE running in USER mode
runmodes: thread stack size of 0 too small: setting to 512k
threshold-config: Error opening file: "C:\Program Files\Suricata\threshold.config": No such file or directory
suricata: setrlimit unavailable.
threads: Threads created -> RX: 1 W: 12 FM: 1 FR: 1 Engine started.
suricata: Signal Received. Stopping engine.
pcap: read 1 file, 15388 packets, 1232460 bytes
C:\Users\Sheheryar Altaf\Desktop\project>
```

Fig. 13: Suricata processing the injected capture - 15,388 packets read, engine ran to completion.



```
fast.log
File Edit View
[11/12/2012-16:03:00.444211] [**] [1:1000003:1] Unauthorized Write Multiple
Coils [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.50.99:59758 ->
141.81.0.46:502
```

Fig. 14: Resulting alert - sid 1000003 ("Unauthorized Write Multiple Coils") fired for source 192.168.50.99 writing to slave 141.81.0.46, confirming the rule works.

## 10. Conclusion

Across baseline analysis, packet-level inspection, and timing checks, the real capture showed no anomaly: all 1,793 write commands came from the expected master to known slaves, addresses and values fell into two consistent clusters, and timing matched normal asynchronous polling rather than a scripted pattern. To confirm the detection approach still works when an anomaly is present, a synthetic unauthorized write was injected and successfully caught by both a manual Wireshark filter and an automated Suricata rule.

This reflects a core weakness of Modbus: the protocol has no built-in authentication, so any device that can reach port 502 can issue commands a PLC will accept as valid. Detection has to rely on network-level context - source IP, expected roles, and protocol fields - rather than anything the protocol itself provides. This is the same gap exploited in real ICS incidents where attackers issued unauthorized commands using legitimate ICS protocols.

Limitations: the capture is a static file rather than live traffic, the anomaly was injected rather than naturally occurring, and the Suricata rule covers only the two write types observed here, not every possible Modbus write command.